# Endogenizing Nature in Production Theory: A Framework of Economic Commensalism

Fernando Souza da Vieira[1] Alexandre Chibebe Nicolella[2]

[1]Faculty of Economics, Administration and Accounting of Ribeirão Preto, University of São Paulo, Brazil

**Abstract**

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**Keywords:** economic commensalism; environmental economics; rational incentives; commons; carbon markets.

## 1. Introduction

The relationship between economic activity and the natural environment has long been a subject of scholarly and social inquiry, particularly with respect to the tension between the productive use of natural resources and ecological sustainability. Economic theory has long treated natural resources as exogeneous stocks, constraints, sources of land rent, externalities, or environmental costs. The problem addressed in this article is more specific: in conventional production theory, nature rarely appears as an endogenous, dynamic, and strategically managed factor capable of affecting future productivity, private incentives, and intertemporal welfare. Although land, capital, and labor are commonly treated as fundamental inputs of production, the ecological systems that sustain them are often

[1] fernandodvieira@usp.br
[2] nicolella@usp.br

modeled as external constraints, fixed stocks, or sources of externalities, rather than as dynamic components of the production process itself (JEHLE; RENY, 2011; STIGLITZ, 1989; DASGUPTA, 2001).

As a result, environmental degradation is often treated as an ex post consequence of production decisions, to be addressed through corrective mechanisms such as taxation (PIGOU, 2010), regulation, property-rights allocation (HARDIN, 1968), carbon pricing, or collective governance arrangements (OSTROM, 2002). Altough these mechanisms have shaped much of modern environmental policy, they often operate outside the agent's core production calculus. In practical terms, this separation is reflected in legal and institutional arrangements that impose preservation duties on private agents without fully integrating ecological preservation into their economic rationale. For example, an economic agent may be required to preserve part of a biome even when doing so reduces the amount of productive land available and, consequently, constrains the agent's production capacity, as occurs in Brazil, where legislation restricts land use to varying degrees according to the biome (BRASIL, 2012). This requirement creates an opportunity cost for agricultural production, since land that could otherwise be used productively must be preserved in the interest of protecting a collective good. The agent therefore faces a trade-off between production and preservation, often framed as a conflict between economic growth and environmental conservation, or a public benefit at the expense of individual private costs.

This dichotomy can be understood through three recurring paradigms of human–environment interaction: predation, mutualism, and extractivism. Predation denotes the unsustainable and often illegal depletion of natural resources; mutualism refers here to a preservation-oriented ideal in which human use is strongly limited in favor of ecological integrity; and extractivism occupies an intermediate position, allowing resource use within supposedly sustainable or regenerative limits (GUDYNAS et al., 2009; ACOSTA, 2013). From an efficiency perspective, however, none of these paradigms fully resolves the production–preservation trade-off. Predation disregards ecological constraints and generates negative externalities, while strict mutualism may excessively restrict productive potential (DASGUPTA, 2001; OSTROM, 1990). Extractivism, in turn, may delay ecological collapse but remains vulnerable to regulatory failure: if too restrictive, it underuses productive resources; if too permissive, it reproduces the overuse problem it was meant to avoid (CHICHILNISKY, 2017). Therefore, natural resource allocation tends to deviate from Pareto efficiency whenever private incentives are not aligned with social and ecological costs, which motivates the search for institutional arrangements capable of reconciling productive efficiency with environmental sustainability (HANLEY et al., 2007).

Against this background, the central **problem** addressed in this article is whether environmental preservation can be theoretically reframed not merely as a constraint on production, but as an endogenous component of economic decision-making. More specifically, the article asks whether private productivity and profitability can be reconciled with socially desirable environmental preservation through incentive-compatible arrangements capable of overcoming the limitations of the prevailing extractivist logic.

The **hypothesis** advanced here is that, when ecological variables are incorporated into production theory and when incentives are designed to reflect spatial, intertemporal, and behavioral constraints, environmental preservation may become compatible with individual rationality rather than dependent solely on coercion or moral commitment. In this sense, the article rests on four **premises**: first, the environment can be represented as an endogenous cofactor of production; second, many existing environmental instruments fail to alter the structural private payoff that drives degradation; third, preservation becomes stable only when it is compatible with individual rationality and strategic equilibrium; and fourth, such compatibility requires governance mechanisms that are locally anchored in ecological units, while remaining subject to broader systems of verification and institutional credibility.

The **objective** of the article is therefore to develop the conceptual foundations of a framework for endogenizing nature in production theory and for reinterpreting preservation as a potentially rational, incentive-compatible, and productivity-enhancing economic strategy.

## 2. The Limits of Current Environmental Governance

—> Mecanismos atuais de governança ambiental (regulação, precificação, mercados de carbono, etc.) são insuficientes para lidar com a complexidade e a dinâmica dos sistemas socioecológicos.

—> Dados e evidências mostram que, apesar dos esforços, as tendências de degradação ambiental e perda de biodiversidade continuam, indicando falhas estruturais nos modelos de governança atuais.

—> Citar esforços internacionais (Acordo de Paris, Objetivos de Desenvolvimento Sustentável) e nacionais (políticas ambientais, leis de proteção) que não têm conseguido reverter as tendências negativas.

—> Usar dados de desmatamento, perda de biodiversidade, emissões de gases de efeito estufa, Brasil e mundo, para ilustrar a insuficiência dos mecanismos atuais.

–> Concluir que é necessário repensar os fundamentos teóricos e práticos da governança ambiental, buscando abordagens que integrem melhor a natureza como parte dos sistemas econômicos e sociais, e que sejam mais adaptativas, inclusivas e eficazes.

## 3. Literature Review: From Externalities to Incentives

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## 4. Model

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# 5. Results

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# 6. Discussion

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# 7. Final Remarks

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